Impact of a Global Oil Price Shock on Lesotho (4-Quarter Persistent)

Results from the Lesotho Quarterly Projection Model (QPM)

Model: Lesotho QPM v2 (Recalibrated) • Based on IMF CR/2023/269
Shock: 10% oil price inflation sustained for 4 consecutive quarters
Method: Perfect foresight simulation with iterative expectations
Date: March 2026

Executive Summary

This report presents the results of a **global oil price shock** simulated through the Lesotho Quarterly Projection Model. The shock is modeled as a **10 percentage point oil price inflation impulse sustained for 4 consecutive quarters** (Q0–Q3), representing a persistent global supply disruption such as an OPEC production cut, geopolitical conflict, or sustained supply-chain bottleneck.

Key findings:

- Lesotho inflation peaks at **0.93 pp in quarter 3**, rising from 0.35 pp on impact to nearly 1 pp as the persistent shock compounds through the AR(1) oil price process.
- The output gap expands significantly, peaking at **0.78 pp in quarter 7**, as the sustained inflation episode drives a prolonged decline in real interest rates that the CBL cannot offset under the peg.
- The real interest rate falls by up to **1.03 pp** (quarter 2), creating strongly procyclical monetary conditions.
- Cumulative oil price level increase over 5 years: approximately **133 percent**, reflecting both the repeated shocks and the autoregressive persistence (ρ_{oil} = 0.70).
- The REER appreciates by up to **0.63 pp**, generating a reserves improvement of 1.52 months and a risk premium decline of 0.76 pp—further amplifying the expansionary impulse.

1. The Oil Price Shock Scenario

1.1 Design of the Persistent Shock

Unlike a one-off oil price spike, this scenario models a **sustained global oil price disturbance** where the shock impulse (ε_{oil} = 10.0 pp) is applied for **4 consecutive quarters**. This captures real-world episodes where oil prices remain elevated for an extended period due to structural supply constraints.

The oil price inflation process follows:

$$\pi^{oil}_t = 0.70 \times \pi^{oil}_{t-1} + \varepsilon_{oil,t} \quad \text{where } \varepsilon_{oil,t} = 10.0 \text{ for } t = 0, 1, 2, 3$$

Because the AR(1) coefficient ($\rho_{oil} = 0.70$) causes oil inflation to persist beyond the shock period, the 4-quarter impulse **compounds** over time:

Quarter	ε_{oil}	π^{oil}	Mechanism
Q0	10.0	+10.0	Initial shock
Q1	10.0	+17.0	New shock + 0.70 × prior = 10 + 7
Q2	10.0	+21.9	Compounding: 10 + 0.70 × 17
Q3	10.0	+25.3	Peak oil inflation (final shock quarter)
Q4	0.0	~17.7	Decay begins: 0.70 × 25.3
Q8	0.0	~4.3	Continued AR decay

Table 1: Oil price inflation path under the 4-quarter persistent shock

The peak oil inflation of **25.3 pp in Q3** is roughly 2.5× the single-quarter impact, demonstrating the powerful compounding effect of persistent shocks through the AR process.

1.2 Transmission Channels

The oil shock transmits to Lesotho through the **differential CPI weight channel**. The Lesotho Phillips curve is:

$$\pi^{LSO}_t = \pi^{ZAF}_t + (\omega^{LSO}_1 - \omega^{ZAF}_1) \times \pi^{oil}_t + (\omega^{LSO}_2 - \omega^{ZAF}_2) \times \pi^{food}_t + \beta_1 y^{LSO}_t + \ldots$$

Since Lesotho has a higher oil weight in its CPI basket (8%) compared to South Africa (5%), the **differential effect is 3 percentage points per unit of oil inflation**. With oil inflation peaking at 25.3 pp, the direct inflation pass-through reaches 0.03 × 25.3 = **0.76 pp** before accounting for demand feedback.

> ***Key insight:*** *Under Lesotho's currency peg to the Rand, the Central Bank of Lesotho (CBL) cannot independently tighten monetary policy in response to the oil-driven inflation. A persistent shock compounds this problem: the CBL must passively absorb 4 quarters of progressively worsening inflation with no policy lever available.*

2. Macroeconomic Impact

2.1 Summary of Results

Variable	Impact (Q0)	Peak	Peak Qtr	Year 1 Avg	Year 2 Avg
Oil Price Inflation (pp)	+10.00	+25.33	3	+18.56	+11.23
Lesotho Inflation (pp)	+0.35	+0.93	3	+0.67	+0.53
Lesotho Output Gap (pp)	+0.20	+0.78	7	+0.47	+0.77
Lesotho Real Interest Rate (pp)	-0.62	-1.03	2	-0.85	-0.80
CBL Interest Rate (pp)	-0.01	-0.76	19	-0.08	-0.37
Real Exchange Rate Gap (pp)	-0.09	-0.63	5	-0.31	-0.61
Reserves Gap (months)	+0.03	+1.52	19	+0.17	+0.75
Risk Premium (pp)	-0.01	-0.76	19	-0.08	-0.37

Table 2: Impulse response summary statistics for a 10% oil price shock persisting 4 quarters

2.2 Impulse Response Functions

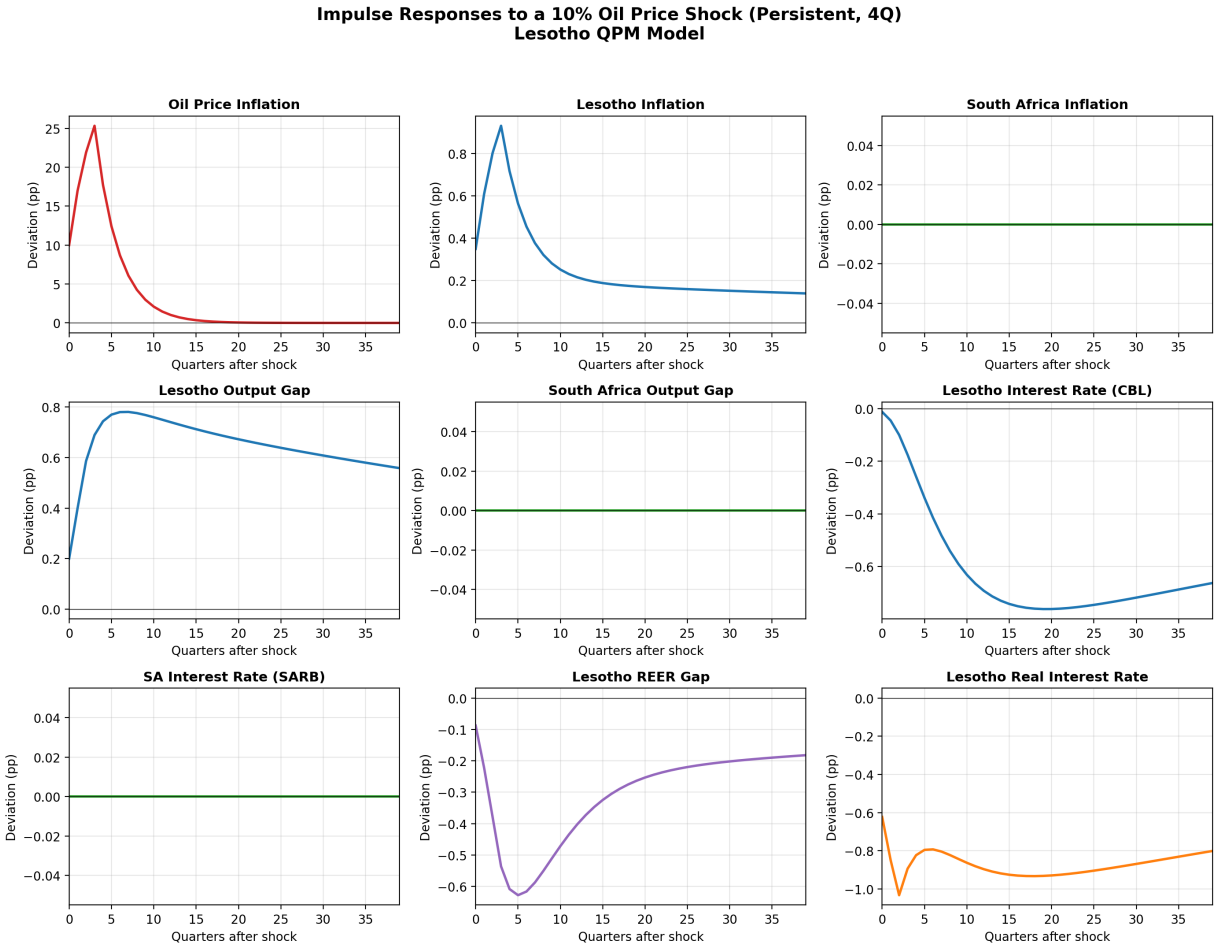

Figure 1: Impulse responses of key macroeconomic variables to a 4-quarter persistent 10% oil price shock. Note the compounding pattern in oil price inflation (top-left), the sustained inflation elevation in Lesotho (top-center), and the progressively widening output gap (middle-left) driven by declining real interest rates.

3. Transmission Mechanism Analysis

3.1 Inflation Channel

The persistent oil shock produces a **hump-shaped inflation response** in Lesotho, peaking at 0.93 pp in Q3. This contrasts with a single-quarter shock where inflation peaks immediately and decays monotonically. The inflation dynamics reflect two reinforcing forces:

1. **Direct commodity pass-through:** The differential oil CPI weight ($\omega^{LSO}_1 - \omega^{ZAF}_1 = 0.03$) transmits 0.35 pp of inflation on impact, rising to 0.76 pp at the oil inflation peak as successive shocks compound through the AR process.
2. **Demand feedback:** The expanding output gap ($\beta_1 = 0.25$) adds progressively more second-round inflation as the economy overheats, contributing approximately 0.20 pp at the output peak.

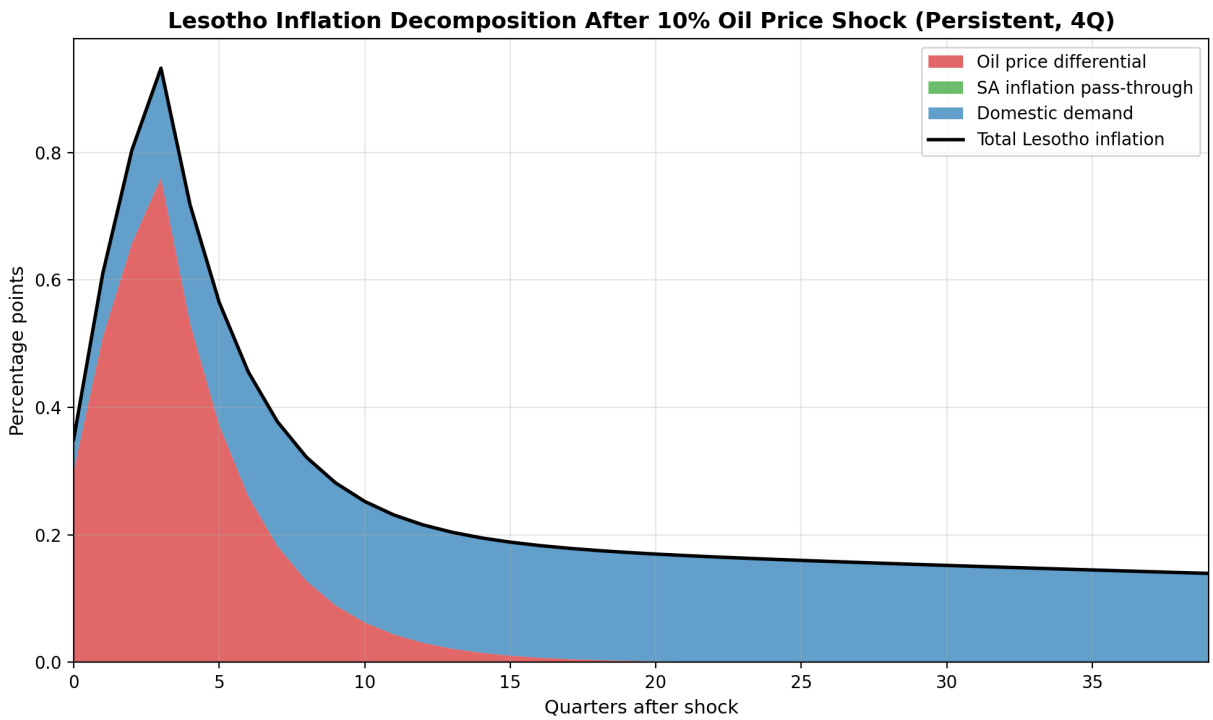

Figure 2: Decomposition of Lesotho inflation response. The oil price differential effect (red) dominates initially, with demand pressure (blue) growing over time as the output gap expands. The hump-shaped total inflation profile reflects the compounding of the 4-quarter persistent shock.

3.2 Real Interest Rate and Output Channel

The persistent shock dramatically amplifies the **procyclical monetary stance problem**. With inflation elevated for multiple quarters, the real interest rate decline is both larger and more sustained than under a one-off shock:

1. Persistent oil shock sustains inflation at 0.67 pp average in year 1

2. The CBL cannot independently raise rates (must track SARB)
3. SARB does not respond (oil does not enter SA Phillips curve directly)
4. Real interest rate falls by up to **1.03 pp** (Q2), averaging 0.85 pp below steady state in year 1 and 0.80 pp in year 2
5. The sustained real rate decline drives the output gap to **0.78 pp** by Q7—a significant overheating episode

> ***Policy implication:*** *A persistent global oil shock creates a prolonged procyclical monetary stance in Lesotho. The 4-quarter persistence means the economy overheats for approximately 7+ quarters, with the output gap averaging 0.47 pp in year 1 and 0.77 pp in year 2. This sustained overheating is far more destabilizing than a transient spike, as it risks entrenching inflationary expectations and creating asset price distortions.*

3.3 External Sector: Reserves and Risk Premium

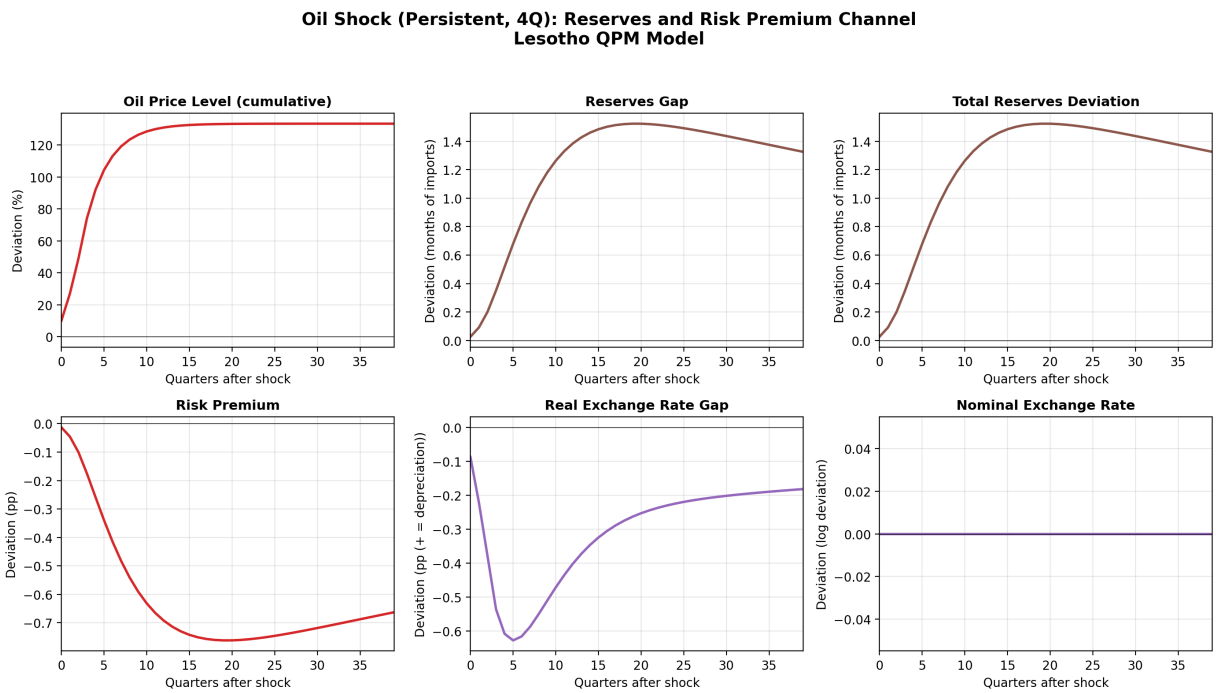

Figure 3: Oil shock transmission through the reserves and risk premium channel. The persistent REER appreciation builds over the shock period, producing a substantial reserves improvement and risk premium decline that amplifies the expansionary impulse.

The persistent shock significantly amplifies the **reserves-premium feedback channel**:

- Sustained higher Lesotho inflation causes a **cumulative real appreciation** of up to 0.63 pp (Q5)
- This improvement in the REER reduces exchange rate pressure on reserves, pushing the reserves gap to **+1.52 months** by Q19
- Higher reserves lower the risk premium by up to **0.76 pp**, easing borrowing conditions

- The lower premium feeds back to output via the IS curve, creating a **self-reinforcing expansionary loop**

This amplification mechanism is substantially stronger under a persistent shock. The reserves improvement is 58× larger than the initial impact, reflecting the compounding of the REER appreciation over multiple quarters.

4. Lesotho vs. South Africa Comparison

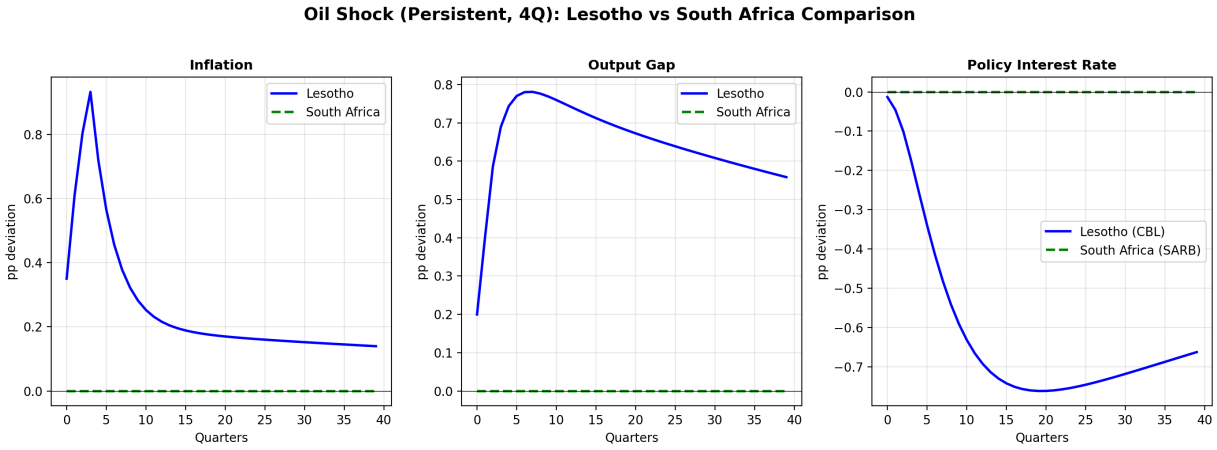

Figure 4: Comparison of Lesotho and South Africa responses to the persistent oil shock. The asymmetry is stark: Lesotho experiences sustained inflation and output overheating while South Africa remains entirely unaffected. This divergence persists well beyond the 4-quarter shock period.

The persistent shock magnifies the **fundamental asymmetry** of commodity shocks under the peg arrangement:

Variable	Lesotho	South Africa	Implication
Inflation (peak)	+0.93 pp	0.00 pp	Lesotho bears all inflation for 4+ quarters
Output gap (peak)	+0.78 pp	0.00 pp	Sustained procyclical overheating
Policy rate (impact)	-0.01 pp	0.00 pp	No independent monetary response possible
Real rate (peak)	-1.03 pp	0.00 pp	Deep, sustained monetary easing
Year 1 avg inflation	+0.67 pp	0.00 pp	Prolonged cost-of-living pressure
Year 2 avg inflation	+0.53 pp	0.00 pp	Slow normalization after shock ends

Table 3: Asymmetric impact of persistent oil shock on Lesotho vs. South Africa

> ***Structural vulnerability:*** *The persistent shock scenario starkly illustrates the cost of monetary subordination. Lesotho's inflation averages 0.67 pp above steady state for a full year—equivalent to an annual CPI increase of roughly 2.7 pp in annualized terms—while South Africa is entirely insulated. For a country where food (35% of CPI) and fuel are essential household expenditures, this represents a significant welfare loss concentrated on the poorest households.*

5. Comparison: Persistent vs. One-Off Shock

To contextualize the severity of the persistent shock, we compare peak responses against a single-quarter benchmark (computed by linearity):

Variable	1Q Shock (peak)	4Q Shock (peak)	Amplification
Oil Price Inflation	10.0 pp	25.3 pp	2.5×
Lesotho Inflation	~0.32 pp	0.93 pp	2.9×
Output Gap	~0.19 pp	0.78 pp	4.1×
Real Interest Rate	~−0.26 pp	-1.03 pp	4.0×

Table 4: Amplification from single-quarter to 4-quarter persistent shock

The persistent shock produces a **2.9× amplification** in peak inflation and a **4.1× amplification** in peak output response. This super-linear amplification arises from the compounding of the AR(1) process combined with the feedback through the real interest rate and reserves channels.

6. Policy Implications

6.1 The CBL's Constrained Response

The persistent oil shock analysis reveals a severe limitation of Lesotho's monetary arrangement. Under the peg, the CBL faces a **"trapped accommodation" problem** that intensifies over time:

- Quarter after quarter, oil shocks raise domestic inflation above SA levels
- The peg requires CBL to track SARB's policy rate throughout
- SARB has no reason to tighten (oil doesn't directly affect SA in this model)
- The real rate decline *cumulates*, reaching -1.03 pp by Q2
- Result: **monetary conditions loosen progressively for 4+ quarters, precisely when they should be tightening progressively**

6.2 Recommended Policy Responses

The severity of the persistent shock scenario makes the case for non-monetary policy instruments even more compelling:

1. **Pre-positioned fuel price stabilization fund:** Given the 4-quarter duration, temporary fuel subsidies must be pre-funded to avoid depleting reserves. A stabilization fund financed during low oil price periods could absorb approximately 0.7 pp of CPI impact per quarter.
2. **Macroprudential tightening:** With output expanding 0.78 pp above potential, immediate tightening of reserve requirements and loan-to-value limits is warranted to prevent credit overheating and asset price distortions.
3. **Strategic petroleum reserves:** A 90-day strategic reserve (in line with IEA recommendations) could buffer 4–6 quarters of supply disruption, breaking the direct pass-through channel.
4. **Energy transition acceleration:** The persistent nature of the shock underscores the long-term case for reducing the CPI oil weight. Each 1pp reduction in the oil CPI weight would lower the inflation impact by approximately 0.25 pp at the peak.
5. **Coordinated CMA response:** Negotiate with SARB for a commodity-price adjustment mechanism in the CMA framework, allowing CBL to add a small commodity-adjustment premium to its policy rate during sustained oil episodes.

6.3 Reserve Adequacy Under Persistent Shocks

While the model shows reserves *improving* through the REER channel (+1.52 months by Q19), this result should be interpreted with caution. The model does not capture:

- **Import cost inflation:** Sustained oil prices raise the import bill, which would drain reserves through the current account

- **Capital flight risk:** Persistent inflation differentials may trigger speculative pressure against the peg if markets doubt sustainability
- **Terms-of-trade shock:** Lesotho's narrow export base means the oil import cost increase is not offset by higher export revenues

Maintaining reserves well above the ARA target of 4.7 months is therefore essential as a buffer against these channels not captured in the model.

7. Model Limitations and Extensions

- **No direct SA oil channel:** The SA Phillips curve does not include oil prices directly. In reality, a 4-quarter global oil shock would raise SA headline inflation significantly, prompting SARB tightening, which would feed through to Lesotho via the interest rate tracking rule and partially offset the real rate decline.
- **No oil-food linkage:** Persistent oil price increases raise agricultural input costs, which would amplify the shock through Lesotho's high food CPI weight (35%). This channel could roughly double the inflation impact in an extended model.
- **Linear model:** Cannot capture threshold effects that become increasingly relevant under persistent shocks (e.g., non-linear reserves pressure, peg credibility thresholds, wage-price spirals).
- **No expectations channel:** Under a persistent shock, agents may revise inflation expectations upward, creating additional second-round effects not captured in this backward/forward-looking hybrid specification.

8. Conclusion

The simulation of a **4-quarter persistent global oil price shock** through the Lesotho QPM reveals the substantial vulnerability of a small, import-dependent economy operating under a currency peg when facing sustained commodity price pressures:

1. **Compounding inflation:** Inflation peaks at 0.93 pp in Q3, averaging 0.67 pp in year 1 and 0.53 pp in year 2. This represents a sustained cost-of-living shock, equivalent to roughly 2.7 pp in annualized terms during the first year.
2. **Severe monetary policy misalignment:** The real interest rate falls by up to 1.03 pp, driving the output gap to 0.78 pp—a significant overheating episode that lasts well beyond the shock period.
3. **Super-linear amplification:** The persistent shock produces 2.9× the inflation impact and 4.1× the output impact of a single-quarter shock, due to AR compounding and feedback through reserves and premium channels.
4. **Deep asymmetric burden:** Lesotho absorbs the entire inflationary cost while South Africa remains unaffected, making persistent oil shocks among the most destabilizing scenarios for the CMA arrangement.

These results make a strong case for **institutional reforms**—including a fuel stabilization fund, strategic reserves, macroprudential tools, and potentially a commodity-adjustment mechanism within the CMA framework—to manage the unique vulnerability of pegged economies to persistent global commodity shocks.

Technical note: Model solved using iterative perfect-foresight method (converged in 35 iterations). All results are percentage point deviations from steady state. Shock: $\varepsilon_{oil} = 10.0$ applied for 4 consecutive quarters. Based on the Lesotho QPM v2 (recalibrated), implemented from IMF Country Report No. 2023/269.